

Student Dropout Prediction & Academic Retention Intelligence System

Problem Statement

Higher education institutions face significant operational and financial challenges with student retention. Identifying at-risk students late in their academic lifecycle leads to unmitigated dropouts. This project establishes an automated predictive engine to evaluate student risk status early based on demographic, academic, and economic markers.

Dataset Overview

The system utilizes the benchmark UCI Student Dropout dataset, containing 35+ input features per student instance. Variables span demographic records, socioeconomic factors, macroeconomic metrics (unemployment rate, inflation, GDP), and detailed 1st and 2nd-semester curricular performance indicators (evaluations, approvals, and grade averages).

Data Preprocessing & Pipeline Architecture

- Raw inputs are structured and validated using **Pydantic** models inside a **FastAPI** backend to ensure type safety.
- Data formatting maps incoming attributes to exact feature names required by the trained model pipeline.
- Preprocessing steps, feature scaling, and transformations are encapsulated within a unified **scikit-learn** pipeline and serialized via joblib.

Exploratory Data Analysis Insights

- Financial indicators—specifically debtor status and tuition fee compliance—exhibit a strong statistical correlation with dropout outcomes.
- Academic throughput metrics, such as the volume of approved curricular units in the first semester, serve as primary predictive drivers for student success.

Model Selection & Training

- Evaluated standard tabular classification models to determine the optimal balance between precision and recall.
- Trained on historical institutional records using cross-validation splits to prevent overfitting and ensure robust generalization on unseen data.
- The finalized model pipeline is saved as `student_dropout_model.pkl` for low-latency inference.

Evaluation Results

- The classification pipeline achieved strong performance across standard evaluation metrics (Accuracy, Precision, Recall, and ROC-AUC).
- The decision threshold was optimized to minimize false negatives for high-risk students, ensuring maximum identification coverage for institutional counseling teams.

Prediction & Production Integration

- Inference is served via a high-performance **FastAPI** REST endpoint (`/predict`) returning numerical predictions, confidence probabilities, and categorized risk statuses.
- Integrated with a multi-tabbed **Streamlit** frontend dashboard that translates complex 35-variable inputs into an intuitive user interface for client-facing presentations.